Latest revision document – JOIV (audit + status + results)

Manuscript: *Attention-Enhanced Dual Autoencoder Clustering for Liver Histopathological Images* (JOIV, decision letters 14 Oct 2026; 5-day window). **Prepared:** 18 Oct 2026. **Package dir:** `joiv_revision/`. All HF repos **private**; nothing submitted or made public.

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1. What has been done (verified)

Infrastructure & reproducibility – DONE

- New isolated work dir `joiv_revision/` (manuscript / response / hf / results).
- Private Hub dataset repo `pytholove/hepar-pannuke-joiv-rev` (33 files: 22 Hepar + 4 PanNuke tar shards + 5 split CSVs + card), model repo `pytholove/dae-kan-attn-joiv-rev` (168 files: `sweep/recon`, `sweep/dec_on`, `sweep/frozen`, code bundle), and a private Collection linking both (see `joiv_revision/hf/HF_MANIFEST.md`).
- gpu-kc (`labkc-comm1`, 2x RTX 3060 12 GB) provisioned: torch 2.8.0+cu128, pytorch-lightning 2.6.6, Hub-authenticated, dataset downloaded + extracted flat (21,168 Hepar + 7,901 PanNuke tiles), code cloned from the Hub bundle.
- Split integrity **asserted in the sweep**: 16,632 tiles/154 specimens train, 4,536 tiles/42 specimens test, `|train ? test| specimens = 0`.

Experiments – DONE (gpu-kc, 3 seeds 42/123/456, 10 epochs, k = 6)

- 21/21 reconstruction runs: 7 variants x 3 seeds, each with test embeddings + biology metadata sidecars (`test_embeddings.npy`, `test_metadata.csv`).
- DEC-objective arm: 3/3 seeds complete (frozen).
- Numbers frozen: `results/frozen_metrics.csv` (72 rows), `frozen_summary.md`, `frozen_stats.csv` (48 paired tests; 0 Holm-significant in full-model families, 1/60 in the variant family).
- k-sensitivity sweep complete (12/12: baseline + full model x k = 3-8): `results/ksens/`. Plateau at k = 4-6, decline at k = 7-8; k = 6 justified as the pre-registered on-plateau choice near the 7-arm treatment structure.
- 20-epoch training-length scout complete (4/4: baseline, kan_only, full, no_eka; seed 42): `results/scout20_verdict.md`. Same ranking as 10-epoch – longer training does not separate the full model.

Writing – DONE (in the .docx)

- `manuscript/JOIV_revised_v1.docx` (4,909 words plain) built from `manuscript_master.md`: abstract 233 words; retired numbers absent by

grep; 38 refs (27 recent, 71%); ELU/CHI equations fixed; specimen-level split; undefined Accuracy/F1 removed; Table II single-protocol profiling; ?III-F frozen-encoder + biology; DEC-instability limitation stated. Now includes the k-sensitivity justification (?II-C) and the 20-epoch scout (?III-A).

- Response letter `response/response_letter_draft.md` filled (no placeholders); comment matrix, reference audit, tables/figures plan staged.

Gap results – banked

Frozen ResNet-50 (treat-AMI 0.245 KMeans / 0.354 GMM), Cramer's V (treatment 0.477 / diet 0.939 / magnification 0.726), single-protocol profiling (baseline 29.4 ms/img, 34.1 fps, 4.9M vs full 37.3 ms/img, 26.8 fps, 5.25M), multi-seed per-magnification control, DEC 128-d latent eval (seed-unstable – reported as limitation, not headline).

2. Verified results snapshot (recon, 3 seeds, standardize_l2, k = 6)

KMeans, held-out test (4,536 tiles). Mean +- SD across 3 seeds:

Model	Silhouette?	DBI?	treat-AMI?	treat-ARI?	diet-AMI?
baseline	0.036+-0.010	5.41+-1.22	0.420+-0.011	0.249+-0.007	0.459+-0.017
bam_only	0.030+-0.009	5.60+-0.85	0.419+-0.001	0.248+-0.003	0.459+-0.002
kan_only	0.024+-0.012	5.98+-0.45	**0.439+-0.007**	**0.289+-0.036**	**0.489+-0.014**
no_bam	0.023+-0.005	5.96+-0.23	0.426+-0.001	0.256+-0.008	0.467+-0.000
no_kan	0.033+-0.007	5.33+-0.19	0.422+-0.002	0.248+-0.002	0.462+-0.004
no_eka	0.039+-0.005	4.94+-0.35	0.416+-0.015	0.243+-0.010	0.454+-0.020
dae_kan_attention	0.036+-0.006	5.32+-0.72	0.433+-0.012	0.279+-0.034	0.478+-0.022

DEC-objective arm (frozen): KMeans Sil 0.040+-0.002, DBI 4.19+-0.20 (recon 5.32), CHI 215+-15 (166), Xie-Beni 5.16+-1.01 (11.44), treat-AMI 0.399+-0.033 (recon 0.433). Evaluation embeddings are the 4096-d bottleneck z for all arms; the 128-d head exists only in DEC training.

20-epoch scout (seed 42): baseline 0.412, kan_only 0.437, full 0.421, no_eka 0.423 – same ranking, no separation from longer training.

Honest reading (drives the manuscript): - **All variants sit within noise of baseline on biological agreement** (0.416-0.439). Only kan_only is nominally ahead. After Holm correction, 1 of 60 modelxmetric cells is significant. - The defensible contribution is (i) the leak-free/biology-validated pipeline, (ii) DEC-objective separability, (iii) confound control – not an internal-index win. No "architecture » baseline" claims.

3. k-sensitivity (KMeans treat-AMI, mean over 3 staged runs per k)

Model	k=3	k=4	k=5	k=6	k=7	k=8
baseline	0.346	0.423	0.427	0.420	0.404	0.392
dae_kan_attention	0.383	0.473	0.455	0.433	0.408	0.396

Plateau k = 4-6 both models; k = 6 pre-registered, not tuned (peaks at k = 4-5).

4. What is left (must-do before submission)

1. **Hub manifest check.** Confirm `sweep/` bulk checksums vs local (`HF_MANIFEST.md` update). Mechanical; bulk already landed (168 files).
2. **DOI resolution.** Verify all 38 refs resolve (conference items need care).
3. **Production figures.** Print-resolution UMAP/montages + JOIV-template tracked-changes copy.
4. **User-owned.** Confirm JOIV portal status (5-day window from 14 Oct may have lapsed – contact editor first if closed), pathologist scoring, final read + submit. Agent stops at package-ready.

5. Risks & blockers

Risk	Status
JOIV 5-day window (-> 19 Oct)	**High** – confirm portal status before submission
Weak multi-seed separation (1/60 Holm-sig)	Framed honestly; DEC + pipeline contribution
No pathologist / SSL suite	Limitation language in manuscript
IMI licence for Hub hosting (even private)	Unconfirmed; repos stay private

6. Deliverables index

- `joiv_revision/LATEST_REVISION.md` (+ `.pdf`) – this document.
- `joiv_revision/response/{comment_matrix,response_letter_draft,reference_audit}.md`

- joiv_revision/manuscript/{manuscript_master.md,manuscript_text.txt,JOIV_revised_v1.docx
- joiv_revision/results/{frozen_metrics.csv,frozen_summary.md,frozen_stats.csv,ksens/,scc
- Hub: pytholove/hepar-pannuke-joiv-rev (dataset), pytholove/dae-kan-attn-joiv-rev (model), Collection pytholove/joiv-revision-dae-kan-liver-histopathology-6ad38ddcc996ae6086 – all private.